

SUMMARY

Full Stack Engineer with 4 years of experience building scalable backend systems and AI-powered platforms. Specialized in Python (Django, FastAPI), React.js, and distributed systems. Hands-on experience designing multi-agent AI systems using LLMs, LangGraph, and Temporal workflows, along with cloud-native architectures on AWS/GCP. Seeking backend / AI-focused roles.

WORK EXPERIENCE

Full Stack Developer - Criterion Networks (Bengaluru, KA) | 1st June 2022 - Present

1) Automated Network Device Onboarding:

- a) Built an end-to-end automated onboarding system to replace the manual process of adding network devices to the platform.
- b) Developed a user-friendly interface using React (MUI) and Django backend to upload and configure qcow2 files (up to 50GB) on Google Cloud.
- c) Implemented resumable file uploads with chunking using Google Cloud Storage APIs to improve reliability and handle large file transfers efficiently.
- d) Automated cloud storage integration and device property configuration, reducing manual effort and onboarding time.
- e) Enhanced system scalability, allowing seamless onboarding of multiple devices with minimal user intervention.

2) AI-Powered Network Automation Platform (MCP):

- a) **Architected an AI-driven network automation platform** using a Python-based MCP server for device management and an intelligent client integrated with a Large Language Model (LLM) to enable network administration via natural language commands.
- b) **Engineered a robust MCP server** leveraging **Netmiko** and **Paramiko** to automate multi-vendor network devices (Cisco IOS, NX-OS), centralizing command execution, configuration, and diagnostics through a secure SSH-based toolset.
- c) **Developed an intelligent client that translates natural language queries** into executable operations by using an AI model to analyze user intent, dynamically select the appropriate remote tool, and resolve hostnames to IP addresses.
- d) **Implemented dynamic device discovery** by creating a system to parse **YAML** configuration files for real-time inventory awareness and build a conversational context memory in the client to provide more accurate, session-aware AI interactions.

3) Agentic AI Orchestration Platform (Multi-Agent AI System):

- a) Built a production-grade multi-agent AI orchestration platform using **FastAPI**, enabling dynamic creation of domain-specific AI agents via JSON configuration without hardcoding.
- b) Designed and implemented a scalable multi-agent workflow system using **Temporal**, ensuring fault-tolerant and durable execution with automatic retries and checkpointing.
- c) Developed a **LangGraph**-based orchestration engine to manage multi-agent conversations, supporting intelligent routing, multi-step execution plans, and context-aware agent collaboration.
- d) Implemented real-time AI chat system using **WebSockets** and Server-Sent Events (**SSE**) for streaming **LLM** responses with reduced latency.
- e) Built dynamic AI agent management system with full **CRUD APIs**, enabling configuration of models, tools, and skills at runtime.
- f) Integrated **MCP** (Model Context Protocol) for dynamic tool discovery and execution, allowing agents to interact with external systems without hardcoded integrations.
- g) Designed multi-tenant architecture with **JWT** authentication, role-based access control, and workspace-level isolation for scalable enterprise usage.

4) Cisco ACI Fabric Automation Platform (ITSM-Integrated Orchestrator):

- a) Built a production-grade automation platform using FastAPI to provision Cisco ACI fabric by integrating with HALO ITSM for change request-driven execution.
- b) Designed end-to-end workflow: ITSM webhook → validation → scheduled task execution (Celery) → APIC provisioning → automated ticket updates and notifications.
- c) Implemented asynchronous task orchestration using Celery + Redis with ETA-based scheduling, retries, and fault-tolerant execution for zero-touch deployments.
- d) Developed provisioning engine using Cisco APIC Cobra SDK to automate tenants, VRFs, EPGs, contracts, L2/L3 networking, and firmware upgrades.
- e) Built audit and compliance system with YAML-based state tracking and payload capture, ensuring full traceability of all infrastructure changes.
- f) Integrated secure authentication (API keys, JWT) and notification system (email + ITSM updates) for enterprise-grade automation workflows.

5) Self-Initiated Tasks:

- a) Shrunk AWS RDS database disks across all environments (staging, beta, production) by implementing blue/green deployment strategies to minimize downtime and ensure smooth transitions.
- c) Optimized 10+ backend APIs by performing query optimizations, enhancing performance, and ensuring only necessary data is served to clients, resulting in faster response times and reduced load on the database through effective Algorithm Design.
- e) **Enhanced the application's user experience** by implementing a platform-wide **Shimmer UI**, replacing traditional loading spinners to reduce perceived load times and provide a seamless content-loading skeleton during data-fetching states.
- f) **Developed a secure backend process** to dynamically **watermark PDF documents** with branding and user-specific information, and automated their subsequent upload to **AWS S3 buckets**, ensuring all generated reports are protected and properly identified

Key Contributions:

- Optimized backend APIs, improving response time by ~30% using advanced ORM techniques.
- Built real-time monitoring and alerting using AWS CloudWatch, reducing debugging time by ~25%.
- Developed serverless workflows (AWS Lambda, GCP Cloud Run) for infrastructure automation and cost optimization.
- Containerized microservices using Docker and implemented CI/CD pipelines via Bitbucket.
- Built real-time notification systems using Django, WebSockets, and Firebase Cloud Messaging.

TECHNICAL SKILLS

Backend: Python, Django, FastAPI, Flask, SQLAlchemy

Frontend: React.js, JavaScript, HTML, CSS

AI/Agents:LLM Integration (Open Router,OpenAI, Anthropic), LangChain, LangGraph, Multi-Agent Systems, MCP, Temporal

Cloud & DevOps: AWS, Docker, CI/CD, Linux

Database: PostgreSQL, MySQL

Tools: Git, Bitbucket, JIRA, Confluence

Methodologies: Agile Methodology, Full Stack Development, Team Collaboration, Problem Solving

PERSONAL PROJECTS

[\[GitHub Link\]](#) ProShop – Full-Stack E-Commerce Platform

Tech Stack: Django, Django REST Framework, React.js, Redux, JWT, Python, JavaScript

- Built a full-stack e-commerce platform from scratch, featuring product search and pagination, a top-product carousel, customer reviews, and a complete shopping cart flow.
- Designed and implemented a secure RESTful API using Django REST Framework to handle all CRUD operations for products, users, and orders, integrating JSON Web Tokens (JWT) for robust user authentication.
- Engineered a dynamic frontend with React.js, managing complex global application state for the shopping cart and user sessions efficiently using Redux with functional components and hooks.
- Built a comprehensive admin dashboard for managing customers, products, and order fulfillment, including functionality to mark orders as delivered and view detailed order information.

EDUCATION

Bachelor of Technology (B. Tech.) National Institute of Technology(tadepalligudem),Andhra Pradesh	June 2017 - May 2021
Senior Secondary-Class XII Sri Chaitanya Junior College	June 2015 - April 2017
Secondary-Class X Santhinikethan school	April 2015